Tommaso Ferrario

MACHINE LEARNING ENGINEER

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Machine Learning Engineer with a passion for building end-to-end systems that turn difficult, real-world problems into workable solutions. Bridging research and production is where the interesting problems tend to live: taking an idea from the paper, figuring out what breaks in practice, and building something robust enough to rely on. The same philosophy applies to mountain hiking, find the best route, adapt when the conditions change, and check the gear before heading into difficult terrain. Good engineering, much like good hiking, is mostly about being prepared for the unexpected.

EXPERIENCE

Machine Learning Engineer — Intesa Sanpaolo, Milan, Italy — Oct 2025 – Present

- Designed and deployed ML pipelines combining OCR and LLM-based models to automate document parsing across financial workflows.
- Integrated models into existing banking systems, building reusable parsing and validation components in Python.

Research Fellow — University of Milano-Bicocca — Sep 2024 – Dec 2024

- Developed a reinforcement learning model for dynamic user profiling in tourism applications, integrated with a recommendation engine (Python, Docker).
- Co-authored a peer-reviewed paper (AINA 2025, Springer) presenting the model architecture and experimental results.

SELECTED PROJECTS

Energy Optimization Platform — Co-founder / Lead Engineer, EORC Ltd (in development)

- Co-building a platform that converts stochastic demand/supply scenarios into multi-stage capacity-planning decisions for energy systems; own end-to-end architecture across optimization, backend, and data visualization.
- Designed a service architecture orchestrated with Docker Compose alongside MongoDB, an MQTT broker, and health-checked optimizer workers.
- Developed an interactive frontend for scenario exploration and result visualization with structured, metadata-driven filtering.

Fine-Grained Food Classification — github.com/TommasoFerrario18/FoodX-251_Classification

- Built a 251-class fine-grained food image classifier (CNN + KNN) with GAN-based restoration for degraded inputs and an object-proposal network.
- Maintained consistent accuracy across varied input quality through a resilient multi-modal pipeline.

TECHNICAL SKILLS

Programming Languages	Python, Julia, Java, MATLAB, SQL
Machine Learning & AI	Deep Learning, Computer Vision, NLP, Reinforcement Learning, PyTorch
Backend & Data	FastAPI, REST APIs, MongoDB, MySQL, PostgreSQL
Infrastructure & DevOps	Docker, Docker Compose, CI/CD, MQTT (Mosquitto), Git, Azure, OpenShift

EDUCATION

MSc in Computer Science — University of Milano-Bicocca — Sep 2023 – Sep 2025

Final grade: 110/110 cum laude. Thesis: *Doob–Meyer Decomposition and AI for Dynamic Scenario Generation*, research conducted at Stevens Institute of Technology, Hoboken, NJ, USA.

BSc in Computer Science — University of Milano-Bicocca — Sep 2020 – Jul 2023

Final grade: 107/110. Thesis: *Automatic traffic analysis: machine learning approaches for the analysis of traffic flows on road networks.*

PUBLICATIONS

Ferrario, T., Fersini, E., Messina, E., Sormani, G. (2025). *Dynamic User Profiling for Personalized Tourism Recommendations Using Reinforcement Learning Models.* Advanced Information Networking and Applications (AINA 2025), Springer. DOI: 10.1007/978-3-031-87781-0_21

AWARDS

Winner, Reply Gen AI competition (2024) — one-day competition, roughly 12 teams. Proposed and prototyped a pipeline that turns a floor plan into a 3D model of the house, then uses generative AI to decorate each room to the occupants' tastes.

LANGUAGES

Italian (native) · English (B2)

INTERESTS

Mountain hiking; Formula 1 (telemetry, simulation, data-driven race strategy).

I authorize the processing of my personal data contained in this CV pursuant to Regulation (EU) 2016/679 (GDPR) and Italian Legislative Decree no. 196/2003 as amended by Legislative Decree no. 101/2018.